

Bonwoo Koo

✉ bonwoo.koo@kaist.ac.kr

📍 bonwoo-koo9.github.io

Research Interest

- Sensor Data Science
- Wearables
- Physical AI

Education

MAR 2026
– Present

Korea Advanced Institute of Science and Technology (KAIST)
M.S in Graduate School of Data Science

AUG 2018 –
FEB 2026

Korea Advanced Institute of Science and Technology (KAIST)
B.S in Department of Industrial & Systems Engineering

AUG 2024 -
DEC 2024

Georgia Institute of Technology
Exchange program at GeorgiaTech Industrial & Systems Engineering in Fall 2024

Experience

JAN 2025
– JAN
2026
(Remote)

Visiting Researcher
GeorgiaTech X Emory Biomedical Engineering, ViTAL Lab (advised by. Hyeokhyen Kwon)

- First-authored a paper on wearable sensor-based detection of neurodegenerative diseases
- Developed an AI-driven framework for analysing multimodal wearable sensor data to detect Mild Cognitive Impairment (MCI) in older adults, using wrist IMU and eye-tracking glasses
- Collected multimodal datasets from 18 older adults (11 with MCI and 7 with Normal Cognition) during a meal preparation task at the Cognitive Empowerment Program, Emory University
- Investigated the association between cognitive decline and both weaker upper limb motor function and delayed eye movements through wrist and eye-tracking data analysis

DEC 2023 –
JUL 2024

Undergraduate Researcher
KAIST Data Science & Artificial Intelligence Lab (advised by. Chanyoung Park)

- Addressed cross-modality bottlenecks in LLM-based Recommendation Framework using cross-attention mechanism and contrastive learning
- Investigated LLM-based RecSys, including both CF-based LLMRec and Multi-Modal LLMRec
- Explored advancements in RecSys from Collaborative Filtering to Side Information-based Filtering
- Explored advancements in Graph based Recommender Systems and Knowledge Graphs

SEP 2022 -
OCT 2023

Undergraduate Researcher
KAIST Financial Engineering Lab (advised by. Woo Chang Kim)

- Co-first authored a paper on deep clustering for financial customer profiling in collaboration with Ph.D. and undergraduate candidates
- Designed a two-stage dimension reduction technique employing Autoencoder, PCA and various Manifold Learning methods, integrated with advanced feature engineering of high-dimensional data
- Led extensive clustering experiments using methods, such as K-means and Hierarchical clustering
- Applied SHAP to enhance cluster explainability and support the design of tailored financial portfolios

Publications

NOV 2025	Visual Accessibility Through Open Shelving: Impacts on Cognitive Load, Motivation, Physical Activity, and User Perception in Older Adults with Mild Cognitive Impairment <i>Journal of Aging and Environment</i> Ibrahim Bilau, Bonwoo Koo, Esther Fu, Wendy Chau, Hyeokhyen Kwon, Eunhwa Yang
DEC 2023	Network-based Exploratory Data Analysis and Explainable Three-Stage Clustering for Financial Customer Profiling <i>Engineering Applications of Artificial Intelligence</i> Insu Choi*, Woosung Koh*, Bonwoo Koo*, Woo Chang Kim

Domestic Conferences

MAY 2023	Improving the Clustering Performance of National Survey of Tax and Benefit Data Using Autoencoder and Dimension Reduction Techniques <i>Korea Intelligent Information Systems Society (KIIS) Conference</i> Insu Choi*, Bonwoo Koo*, Woosung Koh*, Woo Chang Kim
NOV 2022	Review of Reinforcement Learning and Recommender Systems in Finance <i>Korea Institute of Industrial Engineers (KIIE) Conference</i> Insu Choi*, Bonwoo Koo*, Woosung Koh*, Woo Chang Kim

Awards & Honors

MAY 2024	2024 South Korea-US STEM Exchange Program Scholarship <i>Organization: Korea Institute for Advancement of Technology</i> Awarded with \$9,000 scholarship for the Exchange Program in 2024 Fall at GeorgiaTech
AUG 2023	2023 KimYoungHan Global Leader Scholarship <i>Organization: KAIST Scholarship Organization</i>
DEC 2022	2022 NH Investment & Securities Big Data Competition – 3rd Place <i>Title: Persona based Lifetime Portfolio Management via Autoencoders and Deep Clustering</i>

Language & Additional Skills

Languages: Fluent in Korean and English; Conversational Proficiency in Spanish
Technical Skills: Python, Pytorch, Tensorflow

References

Hyeokhyen Kwon

Assistant Professor, Emory University
Computational Behavior and Health Analytics Lab
hyeokhyen.kwon@emory.edu

Woo Chang Kim

Professor, KAIST
Financial Engineering Lab
wkim@kaist.ac.kr

Chanyoung Park

Assistant Professor, KAIST
Data Science & Artificial Intelligence Lab
cy.park@kaist.ac.kr
